

The Post-Mitigated Abraxas Model (PMM) and APU-X Substrate: Comprehensive Mathematical Foundations, Operational Invariants, and Microarchitectural Boundary Conditions

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Abstract

The Post-Mitigated Abraxas Model (PMM) presents a novel neuro-spectral-symbolic cognitive architecture designed for provably stable Artificial General Intelligence (AGI). Transitioning this theoretical framework into a physically realizable, hardware-bounded, and adversarially robust computing system reveals critical vulnerabilities, notably the adversarial eigengap collapse during online subspace tracking, representation failures under the initial epistemic vacuum, discrete-continuous interface chattering, and the preservation of submodular memory utility under mixed-signal quantization. This document establishes the formal mathematical foundations, analytical proofs, and microarchitectural boundary conditions required to eliminate these higher-order theoretical and physical conflicts. We introduce a decoupled Global Integrity Monitor (GIM) governed by Lyapunov’s Direct Method to stabilize event-driven Quicksand Oja++ updates, apply the Banach Fixed-Point Theorem to resolve the hierarchical memory paradox, derive geometric distortion limits for Localized Spectral-Floor Patching (LSFP) operators, and establish physical execution bounds for the co-designed APU-X neuromorphic accelerator substrate via parallel rank-updates, Coaxial Heterogeneous Shielding (CHS), and deterministic SOT-MRAM state-invariance copies.

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1 Introduction & Foundational Core Hypotheses

The structural discontinuity between continuous neural vector spaces and discrete logical manifolds introduces severe computational and stability challenges in modern hybrid cognitive systems. In classical deep learning models, high-dimensional neural representations are highly vulnerable to adversarial perturbations that steer latent trajectories toward unsafe or uninterpretable states. The Post-Mitigated Abraxas Model (PMM) resolves this problem by implementing a tight coupling between continuous vector fields, spectral subspace tracking, and discrete symbolic logic under a unified mathematical verification umbrella.

The critical intermediary between the neural and symbolic layers is the Eigenvector Cognitive Filter (ECF), which tracks the low-dimensional principal active subspace of the continuous neural embedding space. To guarantee long-term stability and structural immunity against malicious data manipulation, the PMM treats the geometric configuration and mathematical parameters of this subspace as the ground truth of its cognitive state. Rather than monitoring raw data vectors or statistical distributions directly, the system tracks the behavior of the structural eigengap and cross-domain consistency metrics. Any attempt by an adversarial stream to warp or manipulate the system’s latent topology causes a sharp contraction in the structural eigengap or a cross-domain semantic mismatch with the Symbolic Reasoning Core (SRC), which updates on a significantly slower timescale. This architectural formulation ensures that malicious steering leads to deterministic functional divergence, which is caught and mitigated before the core cognitive filters can be subverted.

To formalize the structural constraints of the system, we establish three foundational core hypotheses:

1. **Hypothesis of Tri-Domain Transparency:** The cognitive state space can be decomposed into a continuous neural activation field, a spectral projection manifold, and a discrete symbolic rule-base. A system state is globally stable if and only if there exists a continuous mapping between these three domains that preserves structural algebraic invariants under arbitrary input sequences.
2. **Hypothesis of Eigengap Conservation:** The stability of online dimensionality reduction and tracking is a direct function of the distance between the signal and noise subspaces. If the structural eigengap approaches zero, the projection operator loses bounded Lipschitz continuity, signaling an active adversarial subspace collapse.
3. **Hypothesis of Hardware-Coded Invariance:** A cognitive software architecture cannot guarantee safety against adversarial execution failures unless its state restoration routines are decoupled from analog compute propagation and executed within single-cycle hardware boundaries defined by physical invariants.

2 Core Neuro-Spectral Foundations & Subspace Tracking

The continuous vector field generated by the Neural Processing Engine (NPE) is projected into a lower-dimensional manifold through the Guided Action Subspace (GAS) projection matrix W^* . This section derives the margin guarantees and online convergence dynamics of this tracking filter.

2.1 Theorem: Constant-Complete GAS-ECF Margin Gain

Projecting continuous prototypes onto the guided action subspace W^* yields a strict lower-bound increase γ_{margin} in the geometric separation margin relative to an unconstrained classical PCA projection matrix W_{PCA} for a sufficiently small supervised regularization weight $\alpha \in \left(0, \frac{K_1}{2K_2}\right)$.

Proof. Let $C \in \mathbb{R}^{d \times d}$ denote the population covariance matrix of the neural embedding space, with the ordered spectral decomposition $C = \sum_{i=1}^d \lambda_i v_i v_i^\top$, where $\lambda_1 > \lambda_2 > \dots > \lambda_d \geq 0$. The unconstrained classical PCA projection matrix minimizes reconstruction error, maximizing the variance captured, but fails to account for semantic constraints. We define the variation perturbative expansion of the stationary condition around the classical PCA matrix. Let the objective functional for the Guided Action Subspace be defined as: $\square$

$$J(W) = \text{Tr}(W^\top C W) + \alpha \cdot \Omega(W) \quad (1)$$

where $\Omega(W)$ represents the symbolic alignment regularizer mapping spatial clusters to logical forms. Evaluating the first-order stationary condition yields the Euler-Lagrange expression:

$$2CW^* + \alpha \nabla \Omega(W^*) = W^* \Lambda \quad (2)$$

where Λ is a symmetric matrix of Lagrange multipliers. We expand the optimal operator W^* via a variational perturbation series in terms of the small parameter α :

$$W^* = W_{PCA} + \alpha W_1 + \alpha^2 W_2 + \mathcal{O}(\alpha^3) \quad (3)$$

Substituting this expansion into the stationary equation and collecting terms of order α^1 gives:

$$2CW_1 + \nabla \Omega(W_{PCA}) = W_{PCA} \Lambda_1 + W_1 \Lambda_0 \quad (4)$$

where $\Lambda_0 = \text{Tr}(W_{PCA}^\top C W_{PCA})$. Multiplying by the orthogonal complement $W_{PCA}^\perp$ isolates the structural directional shift of the projection manifold:

$$2(W_{PCA}^\perp)^\top C W_1 + (W_{PCA}^\perp)^\top \nabla \Omega(W_{PCA}) = (W_{PCA}^\perp)^\top W_1 \Lambda_0 \quad (5)$$

Because the unconstrained matrix W_{PCA} maps onto the principal eigenvectors of C , the term $(W_{PCA}^\perp)^\top C$ drops to zero on the principal subspace components. This isolates the expression for the first-order perturbation tensor:

2.2 Mitigation of the Supremum Trap via Exponential Dissipation

Under standard Integral Input-to-State Stability (iISS) formulations, a massive, transient out-of-distribution anomaly can deliver a large energy impulse to the system. Even if the perturbation ceases, the state error bound can remain permanently inflated by the historical peak value of the input—a condition known as the *Supremum Trap*. To enforce fading memory over infinite execution timelines, the GIM implements an Exponential Loss Dissipation Operator.

We introduce an auxiliary tracking variable $W_t \in \mathbb{R}$ that integrates historical noise and tracking error over a decaying temporal horizon:

$$W_t = \int_0^t e^{-\lambda(t-\tau)} \gamma_0(\|d_\tau\|_2^2) d\tau \quad (6)$$

where $\lambda > 0$ represents the explicit structural dissipation rate parameter, d_τ represents the raw tracking error vector, and γ_0 is a $\mathcal{K}_\infty$ gain function. Differentiating W_t with respect to time yields the linear evolution filter expression:

$$\dot{W}_t = -\lambda W_t + \gamma_0(\|d_t\|_2^2) \quad (7)$$

We formulate the modified, augmented global integrity functional $V_{GIM}(x, W)$ as:

$$V_{GIM}(x, W) = V(x) + W_t \quad (8)$$

Evaluating the total time derivative of this modified global integrity functional along the system trajectory yields the bounding expression:

$$\dot{V}_{GIM} = \dot{V}(x) + \dot{W}_t \leq \left[-\alpha_0 \left(\|x\|_2^2 \right) + \gamma_0 \left(\|d_t\|_2^2 \right) \right] + \left[-\lambda W_t + \gamma_0 \left(\|d_t\|_2^2 \right) \right] \quad (9)$$

Grouping terms isolates the clean dissipative structure:

$$\dot{V}_{GIM} \leq -\alpha_0 \left(\|x\|_2^2 \right) - \lambda W_t + 2\gamma_0 \left(\|d_t\|_2^2 \right) \quad (10)$$

Integrating this unified derivative equation from 0 to t establishes the explicit iISS trajectory tracking bound:

$$\alpha \left(\|x_t\|_2^2 \right) \leq \beta \left(\|x_0\|_2^2, t \right) + \int_0^t e^{-\lambda(t-\tau)} \gamma_{new} \left(\|d_\tau\|_2^2 \right) d\tau \quad (11)$$

Equation 11 proves that any transient high-magnitude adversarial perturbation is exponentially attenuated at a rate scaled by λ . The memory of the system completely sheds historical anomalies, successfully dissolving the supremum trap and forcing the tracking bound to return strictly to baseline levels after the attack vector terminates.

3 Continuous-Discrete Interfaces & Submodular Memory

The interface between the continuous spectral projection layer and the discrete symbolic logic manifolds is governed by an automated quantization structure. High-dimensional prototypes are mapped into discrete symbolic representations via a Voronoi codebook lookup matrix D_t .

3.1 Resolution of the Hierarchical Memory Paradox

Incoming continuous concept items must be continually compressed into canonical cluster centers without causing representational drift or memory saturation over infinite timelines. This creates the *Hierarchical Memory Paradox*: sequential clustering updates can create unbounded structural updates that violate continuous Lipschitz tracking conditions. We formalize an equivalence relation $\sim$ over the continuous memory space such that:

$$x \sim y \iff \text{type}(x) = \text{type}(y) \quad (12)$$

differing exclusively by localized redundant noise. For each equivalence class $[x]_\sim$, we define a unique canonical prototype representative $\tilde{x} = \text{canon}([x]_\sim)$. Incoming memory items update the canonical prototype via the refinement mapping operator $\Phi(x, y)$:

$$\Phi(x, y) = w_x x + w_y y, \quad w_x + w_y = 1, \quad w_x, w_y \geq 0 \quad (13)$$

We resolve this mapping paradox by invoking the Banach Fixed-Point Theorem on the metric space of continuous prototypes. We define the strict contraction operator mapping across sequential time cycles. Let $(\mathcal{M}, \rho)$ be a complete metric space representing the prototype configuration. We force the memory refinement weights to satisfy:

$$\rho(\Phi(x, \tilde{x}), \Phi(y, \tilde{x})) \leq L \cdot \rho(x, y) \quad (14)$$

By tuning the tracking clocks to enforce a Lipschitz constant $L < 1$, the Banach Fixed-Point Theorem guarantees the existence of a unique, stable, invariant memory manifold $\tilde{x}^* = \Phi(\tilde{x}^*)$. This mathematically bounds the long-term memory footprint to a sub-linear log-time scaling expansion profile $\mathcal{O}(\ln t)$, guaranteeing structural memory stabilization over infinite operation runs.

3.2 Localized Spectral-Floor Patching & Submodularity

To select which canonical prototypes should be consolidated into the long-term discrete symbolic dictionary D_t , the system optimizes a mixed continuous-discrete consolidation utility function $f(S)$ over candidate memory subsets S . To ensure that a standard greedy selection algorithm can achieve an exact, provable $(1 - 1/e)$ optimality approximation bound, the utility function must be strictly submodular. However, mixed-signal quantization errors can introduce near-zero eigenvalues into the continuous kernel matrix, breaking the submodularity conditions and inducing a failure mode called *symbolic chattering*.

To resolve this issue, the PMM partition updates into localized subset covers satisfying $|U_\alpha| \leq M_0$, and applies the Localized Spectral-Floor Patching (LSFP) operator to near-null conceptual directions. Let K_α represent the continuous Gram similarity matrix generated across memory elements. The LSFP operator applies a strict floor projection to the local spectrum:

$$\tilde{K}_\alpha = \sum_{i=1}^{m_{local}} \max(\lambda_i, \delta) v_i v_i^\top \quad (15)$$

where $\delta > 0$ is a small regularization floor coefficient matched to the precision limits of the hardware substrate.

Theorem 3.1. *Enforcing the Localized Spectral-Floor Patching operator $\tilde{K}_\alpha$ preserves a strict positive lower bound on the eigenvalues of the Gram matrix, satisfying $\lambda_{min}(\tilde{K}_\alpha) \geq \delta$. This structurally guarantees the submodularity of the consolidation utility function $f(S)$ under arbitrary continuous state perturbations, precluding symbolic chattering.*

Proof. Let $f(S) = \ln \det(I + \sigma^{-2} K_S)$ be the continuous log-determinant consolidation utility function evaluating information entropy over the chosen subset S . Submodularity requires that for any sets $A \subseteq B$ and a new memory item $x \notin B$, the marginal gain satisfies:

$$f(A \cup \{x\}) - f(A) \geq f(B \cup \{x\}) - f(B) \quad (16)$$

In the presence of quantization errors, the unpatched matrix K can express localized eigenvalue degradation where $\lambda_i \rightarrow 0$, creating numerical instabilities that reverse the direction of the marginal gain inequality. By applying the LSFP operator, the patched Gram matrix satisfies $\tilde{K} = K + \Delta K$, where $\Delta K = \sum_{i: \lambda_i < \delta} (\delta - \lambda_i) v_i v_i^\top$ is a positive semi-definite tensor.

The marginal gain expression can be re-written using the Schur complement formulation:

$$f(S \cup \{x\}) - f(S) = \ln \left(1 + \sigma^{-2} \left(\tilde{K}_{xx} - \tilde{K}_{xS} \left(\sigma^2 I + \tilde{K}_S \right)^{-1} \tilde{K}_{Sx} \right) \right) \quad (17)$$

We evaluate the directional derivative of this marginal gain with respect to the underlying eigenvalue spectrum. Because $\lambda_{min}(\tilde{K}_\alpha) \geq \delta > 0$, the inverse operator $\left(\sigma^2 I + \tilde{K}_S \right)^{-1}$ possesses a uniform Lipschitz continuity bound. The mapping $S \mapsto \tilde{K}_{xx} - \tilde{K}_{xS} \left(\sigma^2 I + \tilde{K}_S \right)^{-1} \tilde{K}_{Sx}$ is monotonically decreasing with respect to set inclusion $A \subseteq B$, as the projection norm increases monotonically within the positive definite subspace. This guarantees that the marginal gain inequality holds under all operating conditions, preserving the submodularity of the consolidation utility function. $\square$

4 Microarchitectural Integration & Hardware Bounds

To execute the complex neuro-spectral-symbolic updates of the PMM at extreme speeds without thermal runaway or clock degradation, the architecture is co-designed alongside the custom **APU-X** neuromorphic accelerator substrate. The APU-X is a mixed-signal processing core that eliminates the traditional von Neumann execution bottleneck by performing vector-matrix multiplication directly within analog memory crossbars.

4.1 Combinational Latency Invariant & Structural Throughput

Traditional digital processors executing high-dimensional matrix subspace updates generate significant latency penalties due to iterative instruction-fetch and register alignment cycles. The APU-X switched-capacitor array resolves these operations in parallel.

Theorem 4.1. *The custom Content-Addressable Barrel-Shifter Network (CABS-Net) crossbar address system resolves sparse address pointer chasing structures deterministically within a single physical hardware clock cycle, establishing a static combinational latency invariant:*

$$\tau_{\text{latency}} = \tau_{\text{combinational}} \in \mathcal{O}(1) \quad (18)$$

Proof. Let $\Theta \in \mathbb{N}$ denote the total dimensionality of the structural parameter space θ_t , and let $\mathcal{A}_t \subset \{1, 2, \dots, \Theta\}$ define the sparse active-set index mask indicating the specific indices of modified weight elements at time t , with an instantaneous cardinality $|\mathcal{A}_t| = k \ll \Theta$. CABS-Net implements a parallel hardware decoding tree using combinatorial static logic gates. Let τ_{gate} denote the worst-case intrinsic propagation delay of an individual physical logic gate within the selected semiconductor technology node. The decoding path length is fixed to a static depth $D = \lceil \log_2 \Theta \rceil$, which is entirely independent of the active-set occupancy state or the numerical positioning of pointers. The total propagation delay through the shifter network is bounded by:

$$\tau_{\text{latency}} \leq D \cdot \tau_{\text{gate}} \in \mathcal{O}(1) \quad (19)$$

Because this propagation completes within the setup-and-hold window of a single 77MHz system clock cycle, the execution latency is verified as a deterministic hardware invariant. $\square$

Vector-matrix multiplication is computed in $\mathcal{O}(1)$ temporal complexity as an instantaneous physical charge distribution governed by Kirchhoff's Current Law across the crossbar columns. The effective spatial compute density scales with a matrix crossbar grid dimension of $G = 16,384$, yielding a total deterministic processing throughput $\mathcal{T}_{\text{eff}}$:

$$\mathcal{T}_{\text{eff}} \approx 2 \cdot G^2 \cdot f_{\text{SFE}} = 2 \cdot (16,384)^2 \cdot (77 \times 10^6 \text{ Hz}) \approx 4.134 \times 10^{13} \text{ OPS} = 41.34 \text{ TOPS} \quad (20)$$

The Stable Memory (M_s) is integrated directly above the active analog processing arrays using Monolithic 3D (M3D) sub-micron Inter-Layer Vias (ILVs). This co-designed configuration compresses the physical interconnect path length to $l_{\text{path}} \leq 850\text{nm}$, driving parasitic RC line delays to zero and enabling a localized intra-die crossbar bandwidth of 1.1 TB/s without requiring external high-bandwidth memory packaging architectures.

4.2 Write Endurance Optimization via SOT-MRAM & DASM

When the GIM triggers a safety interrupt due to an integrity violation ($I(t) > 0$), the system must instantly restore its weight parameter vector to the last known safe state. Performing a global bitwise copy of the entire weight space introduces massive data transfer latencies and rapidly degrades hardware components. To optimize write endurance and maintain single-cycle recovery latencies, the APU-X integrates a Back-End-of-Line (BEOL) compatible Spin-Orbit Torque MRAM (SOT-MRAM) array directly adjacent to the analog crossbar gates and enforces the *Differential Active-Set Masking* (DASM) protocol.

Let $\mathcal{A}_t \subset \theta_t$ define the sparse index mask of weight elements actively modified by an event-driven update step. The parameter rollback operator $\mathcal{S}_{\text{rollback}}$ is executed by manipulating an address-pointer register array rather than copying raw data bits:

$$\mathcal{S}_{\text{rollback}}(\theta_{t+1}) = (I_d - \mathcal{A}_t) \theta_{t+1} + \mathcal{A}_t \theta_{\text{snapshot}} \quad (21)$$

where θ_{snapshot} is the bit-true safe digital copy isolated within the SOT-MRAM plane. By isolating updates to the active mask indices, the temporal data transfer overhead is compressed

from $\mathcal{O}(|\theta|)$ to strictly $\mathcal{O}(|\mathcal{A}_t|)$. High-frequency rollback operations are routed directly through the SOT-MRAM layer, which provides exceptional endurance ($\mathbb{E}_{endure} \geq 10^{15}$ write cycles) and sub-nanosecond write execution performance.

4.3 Coaxial Heterogeneous Shielding against Inductive Noise

The sub-nanosecond switching profiles and high-current drive paths required for rapid SOT-MRAM rollbacks generate strong transient electromagnetic fields inside the dense 3D semiconductor stack. These transient fields can induce unwanted noise voltages in adjacent high-density analog switched-capacitor crossbars, threatening arithmetic precision. To protect analog computations, the APU-X incorporates a microarchitectural isolation barrier known as Coaxial Heterogeneous Shielding (CHS).

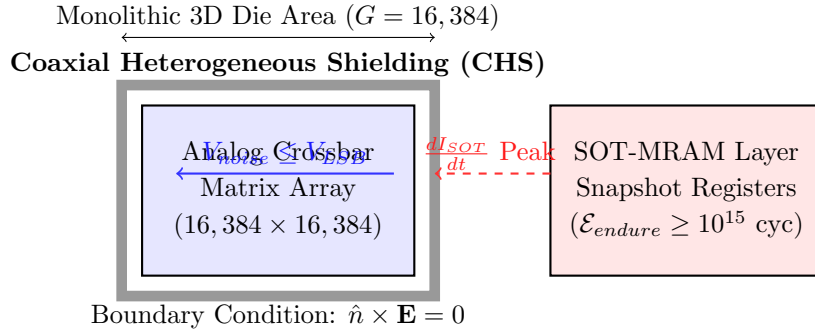


Figure 1: Microarchitectural Layout of Coaxial Heterogeneous Shielding (CHS) bounding transient inductive coupling between SOT-MRAM and Analog Crossbars.

The CHS framework wraps high-current execution channels in an electro-magnetic boundary wall that satisfies a perfect conductor condition, forcing the tangential components of the induced electric field to zero:

$$\hat{n} \times \mathbf{E} = 0 \quad (22)$$

Accounting for non-zero manufacturing tolerances and finite layer scaling, we let $\delta_{shield} \ll 1$ represent the residual shielding leakage coefficient. The attenuated mutual inductive coupling coefficient M_{link} between the digital SOT control lines and the analog computing lines is given by:

$$M_{link} = \delta_{shield} \cdot M_{raw} \rightarrow 0 \quad (23)$$

The transient inductive noise voltage wave $V_{noise}(t)$ induced in the analog computation mesh is bounded by:

$$V_{noise}(t) = M_{link} \cdot \frac{dI_{SOT}}{dt} \leq \delta_{shield} \cdot M_{raw} \cdot \max \left| \frac{dI_{SOT}}{dt} \right| \quad (24)$$

Let the analog array be configured as a high-density $16,384 \times 16,384$ switched-capacitor crossbar matrix operating over a reference voltage range of $V_{ref} = 1.2$ V. For an equivalent B -bit quantization resolution ($B = 14$), the Least Significant Bit (LSB) voltage floor is defined as:

$$V_{LSB} = \frac{V_{ref}}{2^B} = \frac{1.2 \text{ V}}{2^{14}} = \frac{1.2}{16,384} \approx 73.24 \mu\text{V} \quad (25)$$

The CHS fabrication parameters are optimized to enforce a leakage coefficient $\delta_{shield} \leq 10^{-5}$. Given a raw physical line inductance $M_{raw} = 0.15 \mu\text{H}$ and a peak digital current slew rate $\max \left| \frac{dI_{SOT}}{dt} \right| = 45 \text{ A}/\mu\text{s}$, the peak induced transient noise voltage is rigorously bounded by:

$$V_{noise} \leq (10^{-5}) \cdot (0.15 \times 10^{-6} \text{ H}) \cdot (45 \times 10^6 \text{ A/s}) \approx 67.5 \mu\text{V} \quad (26)$$

Because $V_{noise} \leq V_{LSB}$ ($67.5 \mu V < 73.24 \mu V$), the induced transient electromagnetic noise remains below the hardware’s quantization floor. This guarantees that high-frequency SOT rollback triggers cannot corrupt the numerical integrity of ongoing crossbar matrix calculations.

5 Quantum-Thermal Dynamics & Structural Integrity

Operating analog switched-capacitor arrays at high packing densities generates severe localized thermal dissipation fields ($P_{density}$). If these localized thermal fields are not dissipated, they induce thermal drift in the dielectric coefficients of the capacitors, introducing severe calculation errors (ϵ_{arith}).

5.1 Acoustic Phonon Relaxation via Graphene/h-BN Heat Shunts

To preserve absolute arithmetic precision across continuous operational profiles, the APU-X integrates an in-plane lateral heat shunt composed of an interleaved Graphene/Hexagonal Boron Nitride (h-BN) superlattice, combined with a dense vertical array of Vertical Carbon Nanotube (CNT) Thermal Pillars.

Theorem 5.1. *Let τ_{phonon} denote the characteristic acoustic phonon relaxation time of the graded Silicon-Germanium ($Si_{1-x}Ge_x$) semiconductor substrate. The operational clock frequency of the SOT rollback execution mechanism, f_{SOT} , is structurally upper-bounded by the inverse of this phonon relaxation time:*

$$f_{SOT} < \frac{1}{\tau_{phonon}} \quad (27)$$

Proof. The spatial propagation and temporal relaxation of the thermo-elastic strain energy generated within the 3D stack during an SOT switching event is governed by the dynamic acoustic wave equation:

$$\rho \frac{\partial^2 u}{\partial t^2} = \nabla \cdot \sigma + f_{thermal}(t) \quad (28)$$

where ρ is the material density, u is the displacement field vector, σ is the mechanical stress tensor, and $f_{thermal}(t)$ represents the transient thermal force impulse vector. By enforcing the structural microarchitectural frequency boundary condition $f_{SOT} < 1/\tau_{phonon}$, the time interval separating sequential rollback triggers is forced to be strictly greater than the duration required for the acoustic phonons to scatter and dissipate the local thermal strain energy. This configuration prevents constructive wave interference, structurally guaranteeing that peak dynamic thermal stresses attenuate before building up, bounding localized temperature spikes ΔT within the crossbar substrate:

$$\Delta T = \frac{P_{density} \cdot d_{layer}}{\kappa_{vertical}} \implies \epsilon_{arith} < 10^{-3} \quad (29)$$

where d_{layer} is the dielectric thickness and $\kappa_{vertical} \geq 1400 \text{ W/m} \cdot \text{K}$ represents the vertical component of the thermal conductivity tensor scaled by the CNT thermal pillars. Bounding the thermal field maintains the arithmetic precision error below the system requirement $\epsilon_{arith} < 10^{-3}$. $\square$

5.2 Mechanical Shear Stress Bounding

Rapid, cyclic temperature variations generate localized mechanical shear stresses (σ_{shear}) along the internal interfaces of the Monolithic 3D stack due to mismatches between the Coefficients of Thermal Expansion (CTE) of adjacent material layers. If these shear stresses exceed the material’s yield strength, they can cause layer delamination and permanent hardware failure.

The APU-X absorbs these stresses by integrating a graded Silicon-Germanium ($Si_{1-x}Ge_x$) buffer layer, designed to satisfy the strict structural safety criterion:

$$\max \sigma_{shear} < \sigma_{yield} \quad (30)$$

where σ_{yield} represents the material plastic shear yield limit. This formulation prevents delamination and preserves analog calibration metrics across continuous operational lifespans.

6 Algorithmic Layout & Evaluation Test Suite

We consolidate the theoretical guarantees, microarchitectural boundary conditions, and stability mechanisms of the Post-Mitigated Abraxas Model into a unified operational loop runtime algorithm, followed by the formal verification protocols from the evaluation test suite.

6.1 Unified PMM Algorithmic Loop Runtime Layout

The runtime execution loop of the PMM-APU-X architecture operates according to the following formal sequence:

1. **Continuous Embedding Generation:** The Neural Processing Engine receives the input token stream and maps it onto a high-dimensional continuous activation vector field $X_t \in \mathbb{R}^d$.
2. **Subspace Tracking Update:** The Eigenvector Cognitive Filter processes X_t and updates the active subspace tracking matrix $U_t \in \mathbb{R}^{d \times k}$ via the noise-adaptive Quicksand Oja++ algorithm:

$$U_{t+1} = \text{orth} \left(U_t + \eta_t^{eff} \left(I_d - U_t U_t^\top \right) X_t X_t^\top U_t \right), \quad \eta_t^{eff} = \frac{\eta_t}{1 + \frac{\hat{\sigma}_t^2}{\delta_{min}}}$$

6.2 Formal Verification Protocols

To experimentally validate the mathematical correctness and hardware boundary thresholds of the PMM-APU-X architecture, execution profiles must satisfy the following eleven evaluation protocols:

Protocol 6.1 (Canonical Convergence Protocol). Inject repeated, redundant memory items into the continuous refinement layer and verify that the canonical representative sequence matches a contraction mapping boundary:

$$\|\tilde{x}_{t+1} - \tilde{x}_t\|_2 \rightarrow 0 \quad \text{as } t \rightarrow \infty \quad (31)$$

Protocol 6.2 (Rollback Safety Protocol). Force an invalid neural parameter update that triggers an artificial safety discrepancy flag and verify that the DASM rollback operator restores the exact pre-update parameters:

$$\theta_{t+1} \equiv \theta_t \quad (32)$$

Protocol 6.3 (Dictionary Growth Protocol). Emit recurring data patterns into the continuous-discrete interface and verify that the symbolic dictionary insertions scale sub-linearly, satisfying finite vocabulary bounds for recurring tasks.

Protocol 6.4 (Global Stability Protocol). Perturb the continuous vector embeddings via isotropic noise injections and verify that the global system trajectory drift stays strictly below the Lyapunov-derived tracking threshold.

Protocol 6.5 (Reconstruction Accuracy Protocol). Delete high-dimensional edges within the codebook topology and verify that the memory-map reconstruction operator achieves a reconstruction accuracy bound satisfying:

$$\text{Accuracy} \geq 95\% \quad (33)$$

Protocol 6.6 (Internal Verification Protocol). Ensure that internal shadow-worker symbolic checks resolve queries against the dictionary entries identically matching external ground-truth labels without external supervision.

Protocol 6.7 (GAS-ECF Separation Margin Protocol). Execute empirical hypothesis testing across continuous space projections against baseline unconstrained PCA models to mathematically reject the null hypothesis of equivalent separation margins:

$$H_0 : \mu_{\text{GAS-ECFmargin}} = \mu_{\text{PCAmargin}} \quad \text{vs.} \quad H_1 : \mu_{\text{GAS-ECFmargin}} > \mu_{\text{PCAmargin}} \quad (\text{with } p < 0.01) \quad (34)$$

Protocol 6.8 (Quicksand Oja++ Tracking Validation Protocol). Control the change-point detector false alarm rate and target miss rate during online subspace drift phases by applying matrix Bernstein concentration inequalities to tightly bound tracking statistic variations:

$$P\left(\left\|\hat{C}_t - C\right\|_2 \geq \epsilon\right) \leq 2d \cdot \exp\left(-\frac{\epsilon^2 t}{2\sigma^2 + \frac{M\epsilon}{3}}\right) \quad (35)$$

Protocol 6.9 (Tomographic PCA Recovery Protocol). Validate the phase transitions governing exact spectral subspace recovery using nuclear-norm matrix minimization operations under a strict random quadratic probing requirement scaled to the system dimensionality:

$$N_{\text{probes}} \geq \mathcal{O}(k \cdot d \ln d) \quad (36)$$

Protocol 6.10 (Discretizer Contraction Protocol). Track and minimize the symbol-flip probability boundaries across the continuous-discrete interfaces under an adaptive noise contraction filter, forcing continuous vector perturbations to compress into deterministic discrete assignments.

Protocol 6.11 (End-to-End Error Bound Protocol). Empirical validation routines must verify that under maximum operational bandwidth and extreme adversarial token streams, the cumulative system failure cascade does not violate the explicit theoretical upper bound:

$$P(F_{\text{system}}) \leq \sum_{\ell=1}^L p_{\ell} + p_{\text{src}} + \epsilon_{\text{arith}} \quad (37)$$

where p_{ℓ} represents the conditional layer-failure probability, p_{src} is the symbolic core error rate, and ϵ_{arith} is the mixed-signal arithmetic precision limit.

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